Problem Set #2: Understanding Factors Affecting the Early Care and Education Workforce in the United States

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Data for Policy Analysis and Management

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Background

The mental health of early childhood educators is a critical component of a high-quality early care and education system (Murphy & Reid, 2025). Research has shown that educators' psychological well-being affects not only their own work experiences but also children's classroom environments, emotional development, and academic achievement (McLean & Connor, 2015; Oberle & Schonert-Reichl, 2016). In addition, educators with better mental health tend to have more positive interactions with children and report fewer behavioral challenges in their classrooms (Jeon et al., 2014; Kwon et al., 2019). Because early childhood experiences have lasting effects on children's health, educational attainment, and future opportunities, understanding the factors associated with educator mental health is an important research and policy priority (Murphy & Reid, 2025).

In this problem set, you will examine factors associated with depression among staff working in center-based early care and education programs in the United States. Following the analytical approach of Murphy and Reid (2025), you will use data from the 2019 National Survey of Early Care and Education (NSECE) to apply the statistical techniques covered in class and investigate how educator characteristics are associated with depression.

1. Set up working space and import packages

```r
# This cleans up everything in the "Environment" pane.
rm(list=ls())
options("error")
```

```
## $error
## NULL
```

```r
options(lifecycle_disable_verbose_retirement = TRUE)
```

```r
# Import packages
package.list <- c("tidyverse")

# read the `package.list` for the list of packages we will be needing
# Check what packages are installed already
# If any of the packages in the c() list are not currently installed, install them.
# Note: Check the "Packages" tab to verify that they were installed. If they were not, then check the b
for (p in package.list){
  if (!p %in% installed.packages()[, "Package"]) install.packages(p)
  library(p, character.only=TRUE)
}
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
# Set up working directory

## check your directory
getwd()
```

```
## [1] "/home/zacharyjohnigan/UChicago_Student/2_Data_For_Policy_Analys_-_Mgmt/Problem_Sets/Problem_Set
```

```r
## set up working directory
my_directory   <- "/home/zacharyjohnigan/UChicago_Student/2_Data_For_Policy_Analys_-_Mgmt/"
my_data        <- paste0(my_directory, "Data/")
output_dir     <- paste0(my_directory, "Problem_Sets/Problem_Set_2/")
```

2. Load dataset

```r
# Import NSECE 2019 workforce questionnaire
load(paste0(my_data, "37941-0005-Data.rda"))
```

4. Basic cleaning of the data

```r
# Rename dataset
nsece_2019_wf <- da37941.0005
```

Question 1 — Clean and Understand the Data (4 points)

Subset the NSECE dataset to include the following variables:

- `WF9_WORK_WAGE` — hourly wage
- `WF9_WORK_ROLE` — staff role
- `WF9_CHAR_RACE` — race
- `WF9_CHAR_EDUC` — highest level of education completed
- `WF9_CHAR_YEAR_BORN` — staff's birth year
- `WF9_CESD7_TOT` — staff's total score on CESD7 (depression) scale

> **Note:** `WF9_CESD7_TOT` measures the respondent's overall level of depressive symptoms. Scores range from 0, indicating no signs of depression, to 21, indicating severe depressive symptoms. According to the NSECE User Guide (2019), a score of 8 or higher is the recommended cutoff for identifying individuals who may have major depressive disorder. For additional information about this measure, see page 220 of the NSECE User Guide (2019).

Rename the variables using clear, descriptive names. Perform any data cleaning you consider necessary.

```r
# Subset the NSECE dataset and rename the variables

nsece_ps2 <-
  nsece_2019_wf %>%
  select(
    hourly_wage = WF9_WORK_WAGE,
    staff_role = WF9_WORK_ROLE,
    race = WF9_CHAR_RACE,
    education = WF9_CHAR_EDUC,
    birth_year = WF9_CHAR_YEAR_BORN,
    depression = WF9_CESD7_TOT
  )

#glimpse(nsece_ps2)
```

Create the following variables using the code below:

```r
# Create staff age and hourly wage categories

nsece_ps2_recode <-
  nsece_ps2 %>%
  mutate(
    staff_age = 2019 - birth_year,

    staff_age_new = case_when(
      staff_age >= 20 & staff_age <= 40 ~ "Between 20 and 40",
      staff_age >= 41 & staff_age <= 50 ~ "Between 41 and 50",
      staff_age >= 51 ~ "More than 51"
    ),

    staff_age_new = factor(
      staff_age_new,
      levels = c(
```

```r
        "Between 20 and 40",
        "Between 41 and 50",
        "More than 51"
      ),
      labels = c(
        "Between 20 and 40",
        "Between 41 and 50",
        "More than 51"
      )
    ),

    hr_wage_new = case_when(
      hourly_wage <= 10 ~ "Less than $10",
      hourly_wage >= 11 & hourly_wage <= 20 ~ "Between $11 and $20",
      hourly_wage >= 21 & hourly_wage <= 30 ~ "Between $21 and $30",
      hourly_wage >= 31 ~ "More than $31"
    ),

    hr_wage_new = factor(
      hr_wage_new,
      levels = c(
        "Less than $10",
        "Between $11 and $20",
        "Between $21 and $30",
        "More than $31"
      ),
      labels = c(
        "Less than $10",
        "Between $11 and $20",
        "Between $21 and $30",
        "More than $31"
      )
    )
  )

# Examine newly created variables
class(nsece_ps2_recode$staff_age_new)
```

```
## [1] "factor"
```

```r
summary(nsece_ps2_recode$staff_age_new)
```

```
## Between 20 and 40 Between 41 and 50      More than 51              NA's
##              2449               895              1153               695
```

```r
class(nsece_ps2_recode$hr_wage_new)
```

```
## [1] "factor"
```

```r
summary(nsece_ps2_recode$hr_wage_new)
```

```
##       Less than $10 Between $11 and $20 Between $21 and $30       More than $31
##                1088                2359                 335                 209
##                NA's
##                1201
```

Using the variables **role, race, educational attainment, depression, staff age, and hourly wage** — including the new `staff_age_new` and `hr_wage_new` variables — complete the following tasks for **each variable**:

- Identify its level of measurement: **nominal, ordinal, or interval/ratio**.
- Indicate whether it is **discrete or continuous**.
- Present the appropriate **descriptive statistics, tables, or graphs**, including measures of **central tendency, dispersion, and shape** when applicable.
- **Describe and interpret** your findings.

  Full visualizations using `ggplot2` are not required.

Q1 Analysis

Staff Role

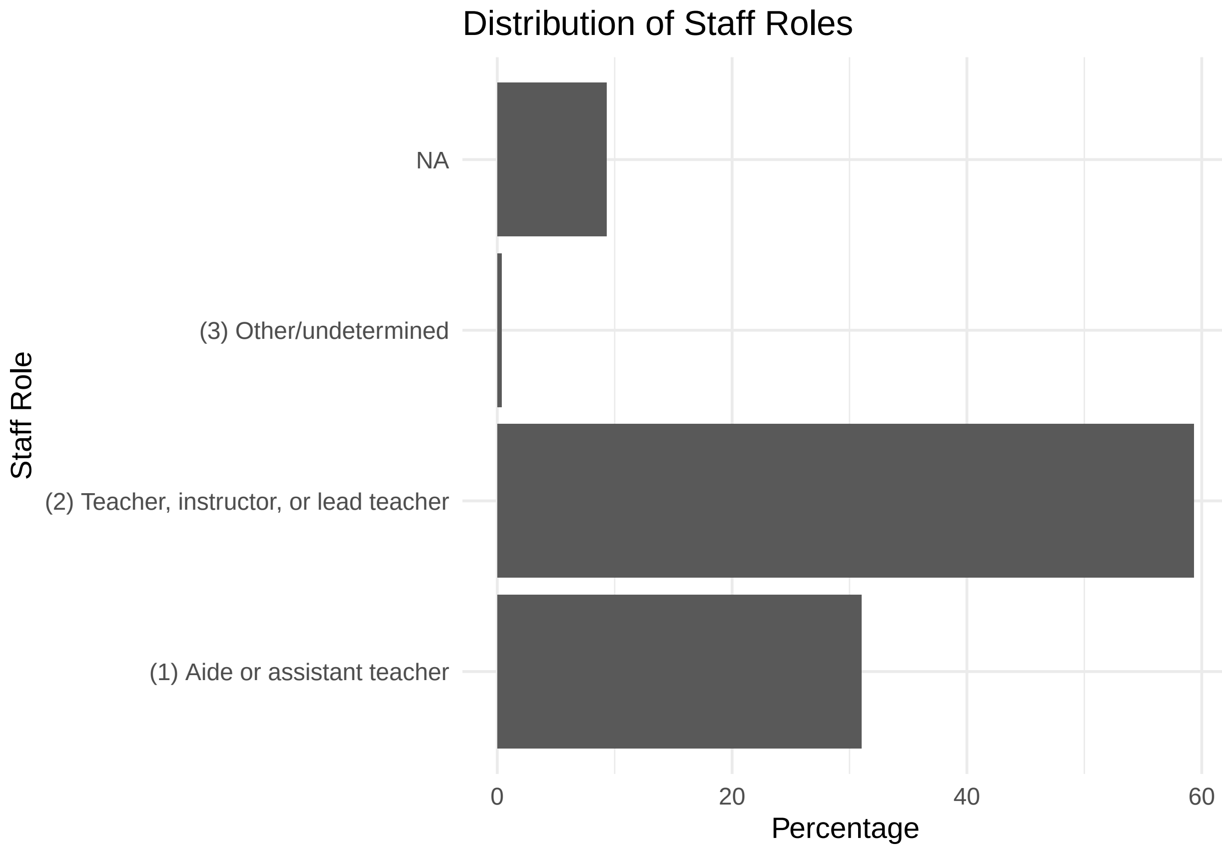
```r
# Variable type: Nominal + Discrete

# Frequency and percentage table
staff_role_table <-
  nsece_ps2_recode %>%
  count(staff_role) %>%
  mutate(proportion = (n / sum(n)) * 100)

staff_role_table
```

```
##                                   staff_role    n proportion
## 1            (1) Aide or assistant teacher 1611 31.0285054
## 2 (2) Teacher, instructor, or lead teacher 3079 59.3027735
## 3                   (3) Other/undetermined   18  0.3466872
## 4                                     <NA>  484  9.3220339
```

```r
ggplot(staff_role_table,
       aes(x = staff_role, y = proportion)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Distribution of Staff Roles",
    x = "Staff Role",
    y = "Percentage"
  ) +
  theme_minimal()
```



Interpretation: Staff role is a nominal, discrete variable because respondents are classified into distinct job categories that have no inherent ranking. Teachers, instructors, or lead teachers were the most common staff role (59.3%), followed by aides or assistant teachers (31.0%). Less than 1% (0.35%) were classified as other/undetermined, while 9.3% of observations were missing staff-role information. The frequency table and bar graph both show that teachers, instructors, or lead teachers make up the majority of respondents.

Because staff role is nominal, the mode is the appropriate measure of central tendency. The modal category is teacher, instructor, or lead teacher. Numerical measures such as the mean, median, standard deviation, and measures of distributional shape are not meaningful for this variable.

Race

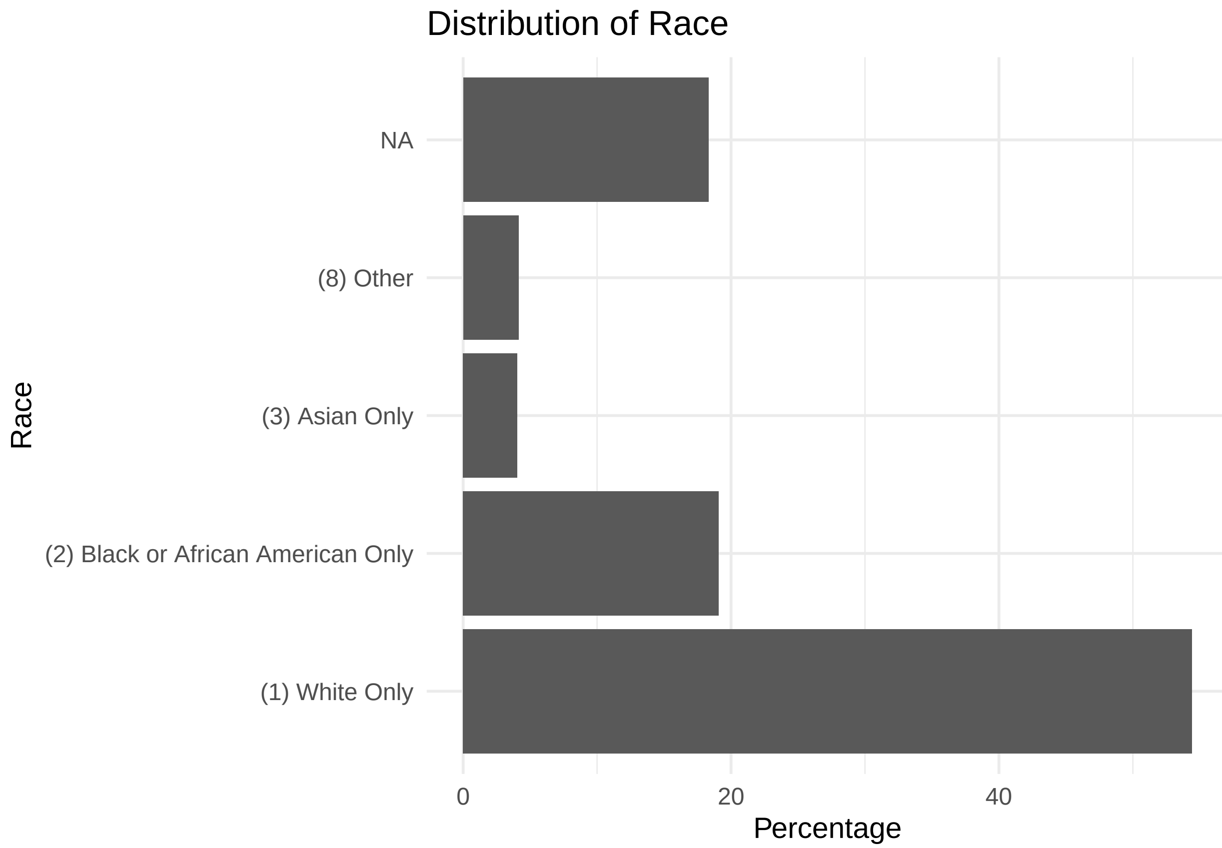
```r
# Variable type: Nominal + Discrete

# Frequency and percentage table
race_table <-
  nsece_ps2_recode %>%
  count(race) %>%
  mutate(proportion = (n / sum(n)) * 100)

race_table
```

```
##                                  race    n proportion
## 1                     (1) White Only 2825  54.410632
## 2 (2) Black or African American Only  991  19.087057
## 3                     (3) Asian Only  210   4.044684
## 4                          (8) Other  215   4.140986
## 5                               <NA>  951  18.316641
```

```r
ggplot(race_table,
       aes(x = race, y = proportion)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Distribution of Race",
    x = "Race",
    y = "Percentage"
  ) +
  theme_minimal()
```



Interpretation: Race is a nominal, discrete variable because respondents are classified into distinct racial categories that have no inherent ranking. White-only respondents were the largest group (54.4%), followed by Black or African American-only respondents (19.1%). Asian-only respondents accounted for 4.0%, and respondents classified as Other accounted for 4.1%. Approximately 18.3% of observations were missing race information.

Because race is nominal, the mode is the appropriate measure of central tendency. The modal category is White Only. Numerical measures such as the mean, median, standard deviation, and measures of distributional shape are not meaningful for this variable.

Educational Attainment

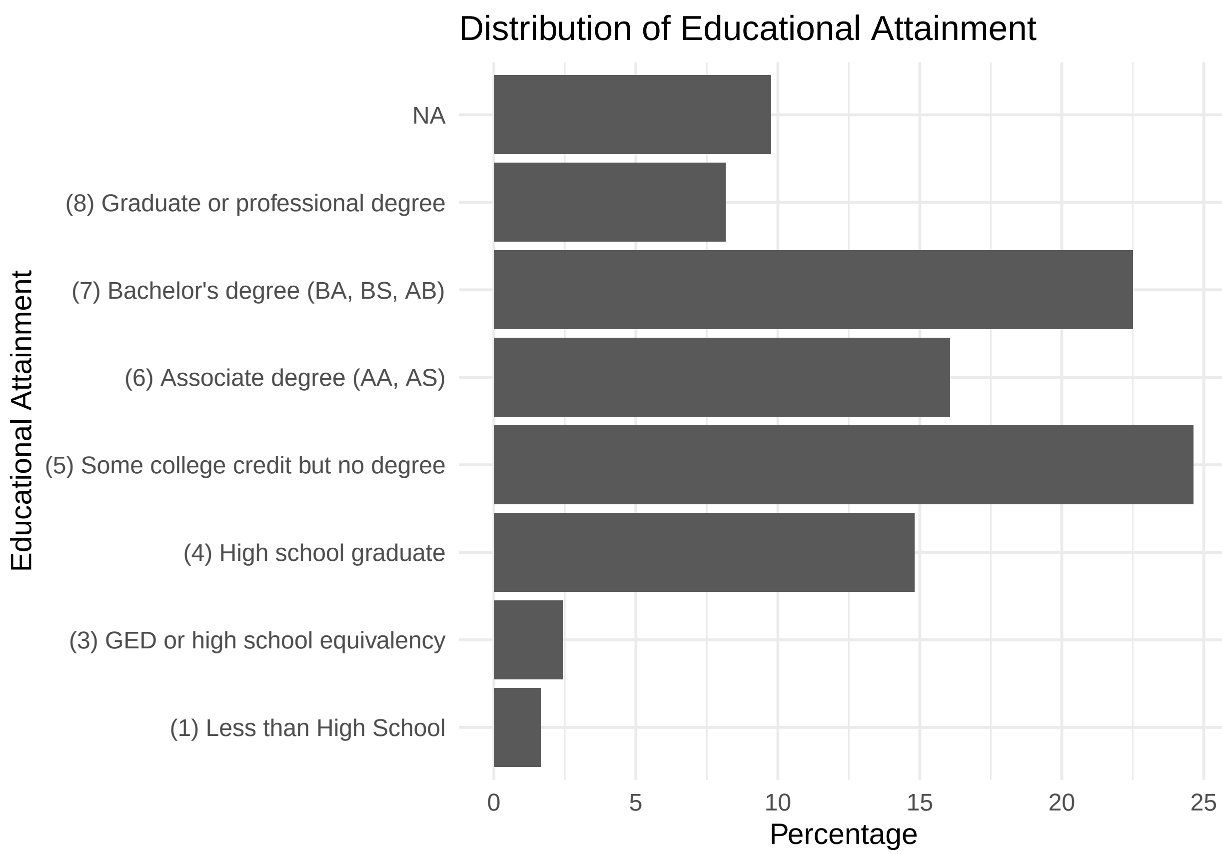
```
# Variable type: Ordinal + Discrete

# Frequency and percentage table
education_table <-
  nsece_ps2_recode %>%
  count(education) %>%
  mutate(proportion = (n / sum(n)) * 100)

education_table
```

```
##                                education    n proportion
## 1             (1) Less than High School   85   1.637134
## 2     (3) GED or high school equivalency  126   2.426810
## 3              (4) High school graduate  769  14.811248
## 4 (5) Some college credit but no degree 1279  24.634052
## 5          (6) Associate degree (AA, AS)  834  16.063174
## 6    (7) Bachelor's degree (BA, BS, AB) 1168  22.496148
## 7   (8) Graduate or professional degree  424   8.166410
## 8                                  <NA>  507   9.765023
```

```
ggplot(education_table,
       aes(x = education, y = proportion)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Distribution of Educational Attainment",
    x = "Educational Attainment",
    y = "Percentage"
  ) +
  theme_minimal()
```



\# Examine ordering of education categories

```
class(nsece_ps2_recode$education)
```

```
## [1] "factor"
```

```
levels(nsece_ps2_recode$education)
```

```
## [1] "(1) Less than High School"
## [2] "(3) GED or high school equivalency"
## [3] "(4) High school graduate"
## [4] "(5) Some college credit but no degree"
## [5] "(6) Associate degree (AA, AS)"
## [6] "(7) Bachelor's degree (BA, BS, AB)"
## [7] "(8) Graduate or professional degree"
```

Cumulative percentage to identify the median category

```
education_median <-
  nsece_ps2_recode %>%
  filter(!is.na(education)) %>%
  count(education) %>%
  mutate(
```

```r
    proportion = (n / sum(n)) * 100,
    cumulative_proportion = cumsum(proportion)
  )

education_median
```

```
##                                 education    n proportion cumulative_proportion
## 1             (1) Less than High School   85   1.814301              1.814301
## 2    (3) GED or high school equivalency  126   2.689434              4.503735
## 3               (4) High school graduate  769  16.414088             20.917823
## 4 (5) Some college credit but no degree 1279  27.299893             48.217716
## 5          (6) Associate degree (AA, AS)  834  17.801494             66.019210
## 6    (7) Bachelor's degree (BA, BS, AB) 1168  24.930630             90.949840
## 7   (8) Graduate or professional degree  424   9.050160            100.000000
```

Interpretation: Educational attainment is an ordinal, discrete variable because respondents fall into distinct educational categories that have a meaningful order. Among respondents with valid education data, the most common category (mode) was some college credit but no degree (27.3%), followed by a bachelor's degree (24.9%). The median category was associate degree (AA, AS) because the cumulative percentage first exceeded 50% at that category. Approximately 9.1% of respondents held a graduate or professional degree, while relatively few had less than a high school education (1.8%) or a GED/high school equivalency (2.7%).

Because educational attainment is ordinal rather than interval/ratio, the mean and standard deviation are not appropriate measures. The median, mode, frequency distribution, cumulative percentages, and bar graph appropriately describe its central tendency and distribution.

Depression

Interpretation:

Staff Age

Interpretation:

Hourly Wage

Interpretation:

Question 2 — Prepare the Data for Linear Regression (2 points)

Create the derived variables needed to estimate a multiple linear regression model:

- Create **k - 1 binary variables for staff role** (`k` = number of categories). After examining the variable, choose a **reference group**.
- Create **k - 1 binary variables for race**. After examining the variable, choose a **reference group**.
- Create **k - 1 binary variables for educational attainment**. After examining the variable, choose a **reference group**.
- Create **k - 1 binary variables for staff age**. After examining the variable, choose a **reference group**.
- Create **k - 1 binary variables for hourly wage**. After examining the variable, choose a **reference group**.

Staff Role Binary Variables

Reference group:

Race Binary Variables

Reference group:

Educational Attainment Binary Variables

Reference group:

Staff Age Binary Variables

Reference group:

Hourly Wage Binary Variables

Reference group:

Question 3 — Estimate the OLS Regression Model (4 points)

Estimate an OLS regression model to examine factors associated with depression among center-based early care and education staff.

The model specified in the assignment is:

$$\text{Depression} = \beta_0 + \beta(\text{hourly wage}) + \beta(\text{staff role}) + \beta(\text{staff age}) + \beta(\text{education level categories}) + \beta(\text{race categories}) + \varepsilon$$

Estimate the Model

```
# Estimate the OLS regression model here.
```

Interpret Statistically Significant Coefficients

Interpret the coefficient estimates for each statistically significant independent variable.

Answer:

Interpret Overall Model Fit

Interpret the overall model fit, including $\mathbf{R^2}$ and **adjusted** $\mathbf{R^2}$.

Answer:

Conclusions

What conclusions can you draw about the factors associated with depression among center-based early care and education staff?

Answer:

Diagnostic Plots

Create:

1. A **residual plot**
2. A **histogram of the model residuals**

```
# Residual plot
```

```
# Histogram of residuals
```

Do these diagnostic plots suggest that the assumptions of **linearity, homoscedasticity, and normality** are reasonably met? What hypotheses do you have about why the plots look the way they do?

Answer: